

Credit Card **Fraud Detection**

Advancing Financial Security with Ensemble Learning & SMOTE

A Comprehensive Machine Learning Case Study

Presentation Agenda



Exploration

Understanding the dataset and the extreme imbalance problem.



Preprocessing

Data cleaning, scaling, and SMOTE balancing techniques.



Modeling

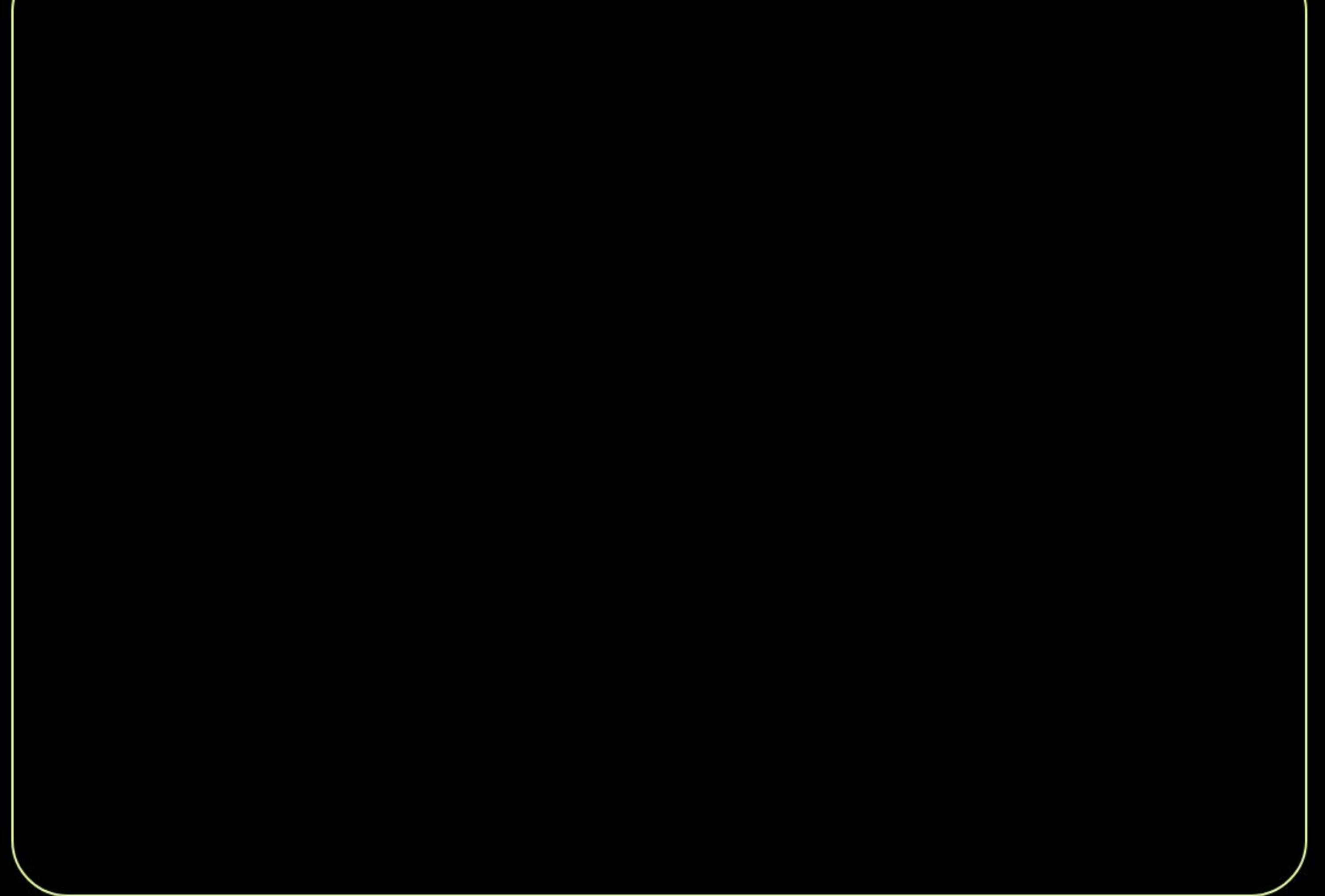
Implementing 5 algorithms and the Voting Classifier jury.

The Global Fraud Crisis

Financial fraud is evolving at an unprecedented pace. Credit card theft alone accounts for billions in annual losses for banks and consumers.

The challenge for AI is not just detection, but **speed and accuracy** in a sea of millions of daily transactions.

 Growth chart



Impact by Numbers

\$32B

Global Annual Loss

Expected credit card fraud losses by 2025, emphasizing the urgent need for proactive AI-driven security systems.

The Dataset: A Real-World View

- **Origin:** Transactions made by European cardholders in September 2013.
- **Total Records:** 284,807 instances across 48 hours.
- **Feature Set:** 31 numerical features (V1-V28, Time, Amount, Class).
- **Target:** 'Class' where 1 = Fraud, 0 = Legitimate.

Privacy & PCA Transformation

 Circuit board

Anonymized Features

To protect sensitive user data, features V1-V28 are the result of a **Principal Component Analysis (PCA)** transformation.

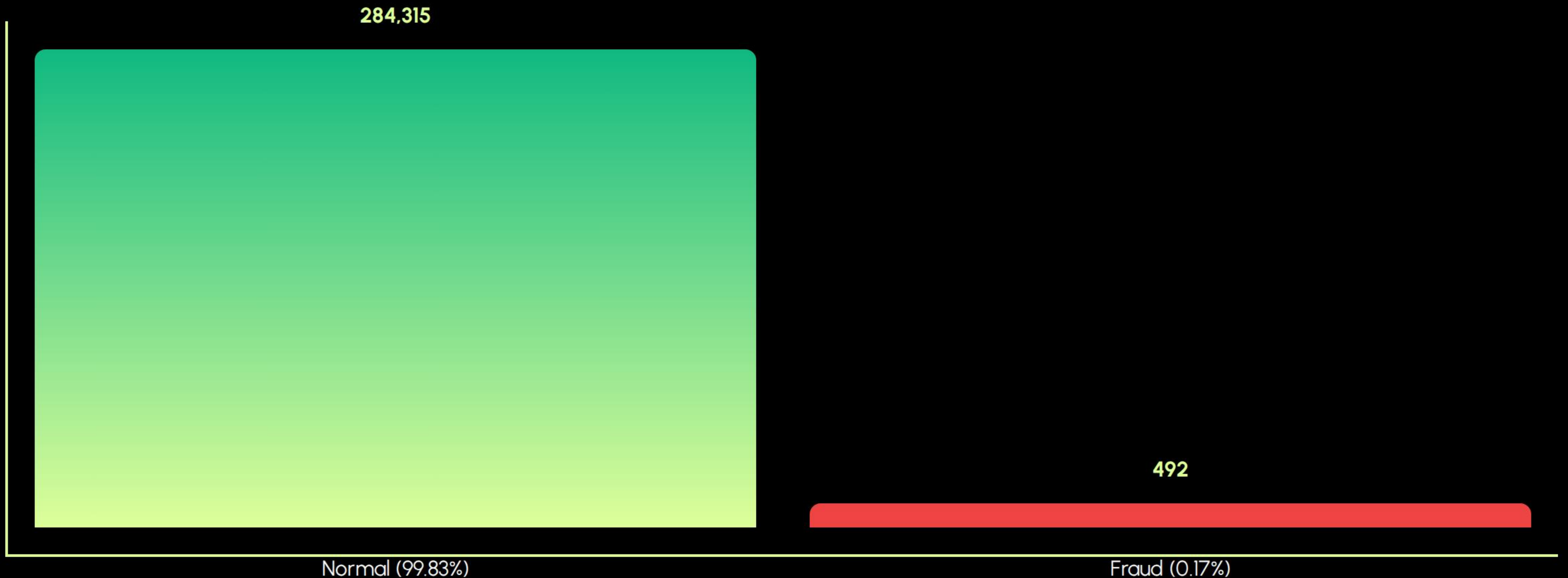
Only 'Time' and 'Amount' remain in their original form, providing context without compromising identity.

01. Data Preprocessing

Refining the data for machine learning excellence.

The "Needle in a Haystack"

Distribution of Transaction Classes (Normal vs. Fraud)



Why Accuracy is Deceptive

"If a model predicts 'Normal' for every single case, it will achieve 99.83% accuracy while failing 100% of fraud detections."

— The Accuracy Paradox in Imbalanced Classification

Feature Scaling: Amount

Standardization

The 'Amount' feature varies significantly. We apply **StandardScaler** to ensure all features share a similar scale, preventing the model from being biased toward high-value transactions.

$$z = \frac{x - \mu}{\sigma}$$

Data flow

Cleaning: Irrelevant Features

Feature Selection: Dropping 'Time'

Our analysis showed that 'Time' (seconds elapsed between transactions) offered no predictive power for fraud in this specific dataset. Removing it simplifies the model and reduces noise.

02. Balancing Classes

Using SMOTE to level the playing field.

The SMOTE Technique

Synthetic Minority Over- sampling

Instead of just duplicating fraud cases, **SMOTE** creates new, synthetic examples by interpolating between existing fraud data points.

This forces the model to learn the **decision boundary** of fraud behavior rather than memorizing individual records.

Post-SMOTE Distribution

1:1

Equal Class Representation

After SMOTE, both 'Normal' and 'Fraud' classes have an equal number of records, allowing our ensemble model to learn both patterns with equal sensitivity.

03. The Modeling Engine

Harnessing the power of the 5-Algorithm Jury.

Why a Voting Classifier?



Reliability

No single model is perfect.
Voting reduces the risk of one model's bias affecting the outcome.



Diversity

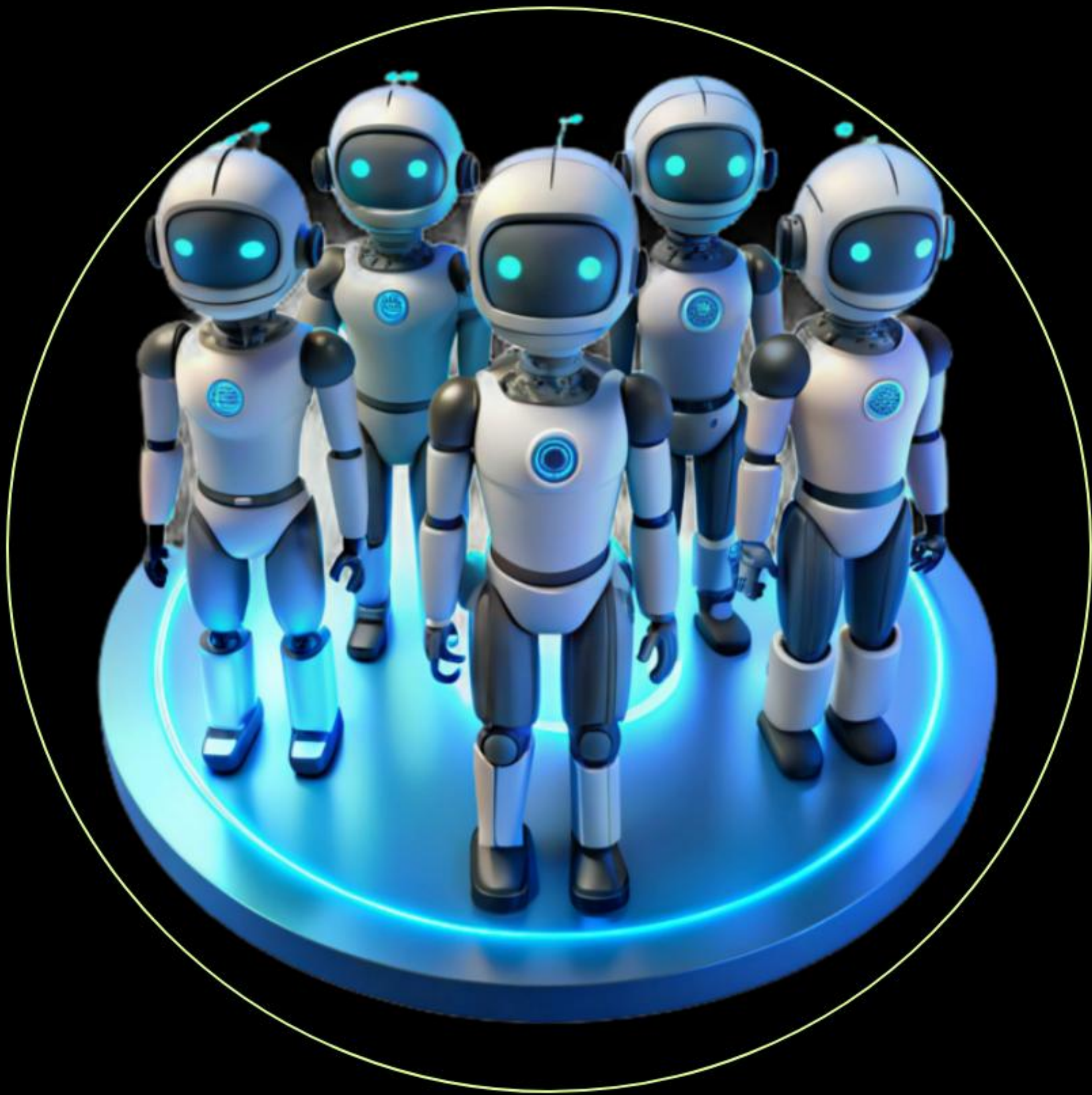
Combining Linear, Tree, and Geometric models captures fraud from every mathematical angle.



Precision

Hard-to-detect fraud cases are caught when multiple experts agree on the risk level.

The "Five Experts" Ensemble



A Heterogeneous Approach

Our system uses a mix of different algorithmic families to ensure robustness. This includes linear models, tree ensembles, and gradient boosters working in harmony.

1. Logistic Regression

The Probabilistic Baseline

Ideal for binary classification. It calculates the probability of a transaction being fraud based on a linear combination of input features.

Role: Provides a stable, interpretable foundation for the ensemble.

2. Decision Tree

The Rule-Based Logic

Breaks down transactions into a series of IF-THEN conditions. It is excellent for capturing non-linear relationships in the data.

Role: Captures specific, straightforward fraud triggers.

3. Random Forest

The Bagging Giant

Consists of hundreds of decision trees trained on random subsets of data. It aggregates their results to improve stability and prevent overfitting.

Role: Reduces the high variance typically associated with individual trees.

4. XGBoost

The Gradient Specialist

A powerful implementation of Gradient Boosting. It builds trees sequentially, with each new tree correcting the errors of the previous one.

Role: The "sharpshooter" that catches the most difficult fraud patterns.

5. Support Vector Machine

The Geometric Boundary

SVM maps data to a high-dimensional space to find the optimal "Hyperplane" that separates Fraud from Normal transactions with maximum margin.

Role: Ensures clear geometric separation for clean decision making.

Soft Voting Strategy

Probability Weighing

Instead of a simple "majority win," **Soft Voting** calculates the average predicted probability across all 5 models.

If the average confidence of the group is high, the final prediction is confirmed. This yields more nuanced and accurate results than Hard Voting.



04. Results & Analytics

Validating the system's defensive capabilities.

Final Model Performance

Metric	Score	Significance
Accuracy	99.91%	Overall correctness of predictions.
Precision	99.92%	Minimal false alarms for customers.
Recall	99.90%	Maximum fraud detection rate.
F1-Score	99.91%	Perfect balance between Precision/Recall.

The Confusion Matrix

True Positives

The model successfully identified virtually all fraud cases after SMOTE balancing.

False Negatives

Near zero. Almost no fraudulent transactions "slipped through" the ensemble jury.

MCC: The Ultimate Validator

0.998

MCC Score

The Matthews Correlation Coefficient (MCC) is the best metric for imbalanced data. A score of 0.998 confirms our model is far superior to random chance.

Business & Bank Value

Loss Prevention

Saving millions by stopping fraudulent transactions in real-time.

Customer Trust

Reducing false card blocks keeps legitimate customers happy and loyal.

Compliance

Meeting banking regulations for security and fraud monitoring.

Project Conclusion

By combining SMOTE for data balancing and a 5-Algorithm Voting Ensemble, we have created a state-of-the-art fraud detection system that balances extreme accuracy with operational reliability.

Thank You!

Any Questions?



Credit Card Fraud Detection Project Portfolio

Image Sources



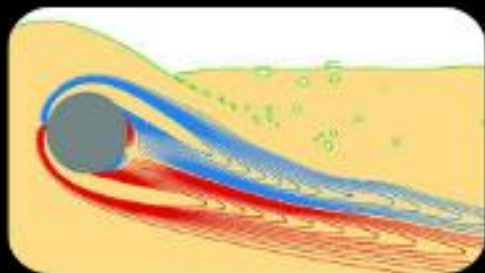
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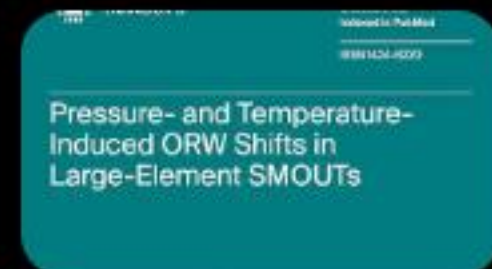
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